Notation: Given a tuple of skew shapes ν:

(i) $\gamma(\nu) :=$ sequence of lengths of diagonals in ν (in increasing reading order)

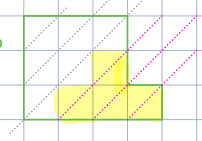
(ii) $n'(\gamma) := \sum_i \binom{\gamma_i}{2}$ (for any γ, but chiefly used for $n'(\gamma(\nu))$)

(iii) $p(\nu) := \sum_{\substack{D \\ D\text{:diag of } \nu}} \mathbb{1}(D \text{ does not contain the first box in a row}) \cdot |D|$ = sum of cells in all diagonals not containing the 1st box of any row.

magic number

(iv) $A(\nu) :=$ # attacking pairs in ν.

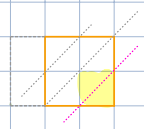
eg. (i) $\nu = (4,3,3)$



$\gamma(\nu) = (1,2,3,2,1,1)$, $n'(\gamma(\nu)) = \binom{1}{2} + \binom{3}{2} + \binom{3}{2} + \binom{2}{2} + \binom{1}{2} + \binom{1}{2} = 5$, $A(\nu) = 0$ (as there is only one shape)

$p(\nu) = 4$

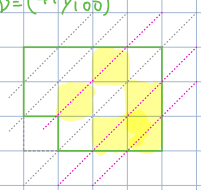
(ii) $\nu = (32/10, 33/11)$



$\gamma(\nu) = (1,1,1,1,2,1,1)$, $n'(\gamma(\nu)) = \binom{1}{2} \times 6 + \binom{2}{2} = 1$, $A(\nu) = 7$

$p(\nu) = 3$

(iii) $\nu = (444/100)$



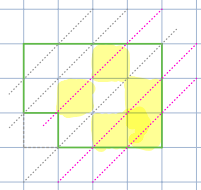
$\gamma(\nu) = (1,2,2,3,2,1)$, $n'(\gamma(\nu)) = \binom{1}{2} + \binom{2}{2} + \binom{2}{2} + \binom{3}{2} + \binom{2}{2} + \binom{1}{2} = 6$, $A(\nu) = 0$

$p(\nu) = 5$

(iv) $\nu = (444/100, 11)$

$\gamma(\nu) = (1,2,1,3,1,2,1)$, $n'(\gamma(\nu)) = 6$, $A(\nu) = 9$

$p(\nu) = 5$



Lemma 7.2.2: For each diagonal D in ν, let D_+ denote the (possibly empty) adjacent diagonal SE to D, i.e. $\tilde{c}(\text{cell in } D_+) = \tilde{c}(\text{cell in } D) + 1$. Then

$$n'(\gamma(\nu)) + p(\nu) = \sum_{\substack{D \\ D\text{:diag of } \nu}} |\{(d,e) \in D \times D_+ \mid d \prec e\}|.$$

Proof: Let $d \in D$.

If $\text{south}(d) \in \nu$, then $(d,e) \in D \times D_+$ with $d \prec e$ iff $(\text{south}(d), e) \in D_+ \times D_+$ (with $\text{south}(d) \prec e$)

If $\text{south}(d) \notin \nu$, then D_+ does not contain the 1st box of any row

$$\therefore \sum_{\substack{D \\ D\text{:diag of } \nu}} |\{(d,e) \in D \times D_+ \mid d \prec e\}| = \sum_{\substack{D \\ D\text{:diag of } \nu}} \Big(\underbrace{\sum_{\substack{d \in D \\ \text{south}(d) \in \nu}} |\{(\text{south}(d), e) \in D_+ \times D_+ : \text{south}(d) \prec e\}|}_{\text{pairs of boxes in } D_+} + \underbrace{\sum_{\substack{d \in D \\ \text{south}(d) \notin \nu}} |\{(d,e) \in D \times D_+ \mid d \prec e\}|}_{\text{Hence this is just } |D_+| \text{ where } D_+ \text{ does not contain the first box in any row}} \Big)$$

note: these counts any diagonal other than the first diagonal

Then d must be the 1st box of D in reading order (i.e. D only has at most one such d)

note that even though D_+ does not consider the first diagonal, it is fine b/c the first diagonal must contain the 1st box in the top row.

$$= \sum_{i \geq 2} \binom{\gamma_i}{2} + p(\nu) = n'(\gamma(\nu)) + p(\nu).$$

b/c D_+ does not contain the first diagonal

b/c $\gamma_1 < 1 \Rightarrow \binom{\gamma_1}{2} = 0$. ($\gamma_1 = 0$ if $\nu = \emptyset$)